

PLAXIS

3D Back-Analysis of a O-Cell Pile Load Test

PLAXIS 3D 2024.2

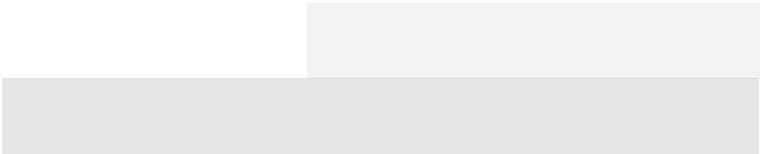


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1.

Introduction

A pile load test is a reliable method to validate pile design and assess uncertainties in the in-situ soil conditions and parameters. One particularly innovative approach is the Osterberg Cell (O-cell) test. Unlike conventional methods that apply load from the top, the O-cell technique applies load from the bottom of the pile. A hydraulically actuated cell, installed near the base, expands under pressure, pushing upward against the shaft and downward against the pile base. The upward resistance mobilizes shaft friction, while the downward force mobilizes end bearing resistance. The pressure is gradually increased until the pile reaches its ultimate capacity, either through side shear, end bearing, or a combination of both.

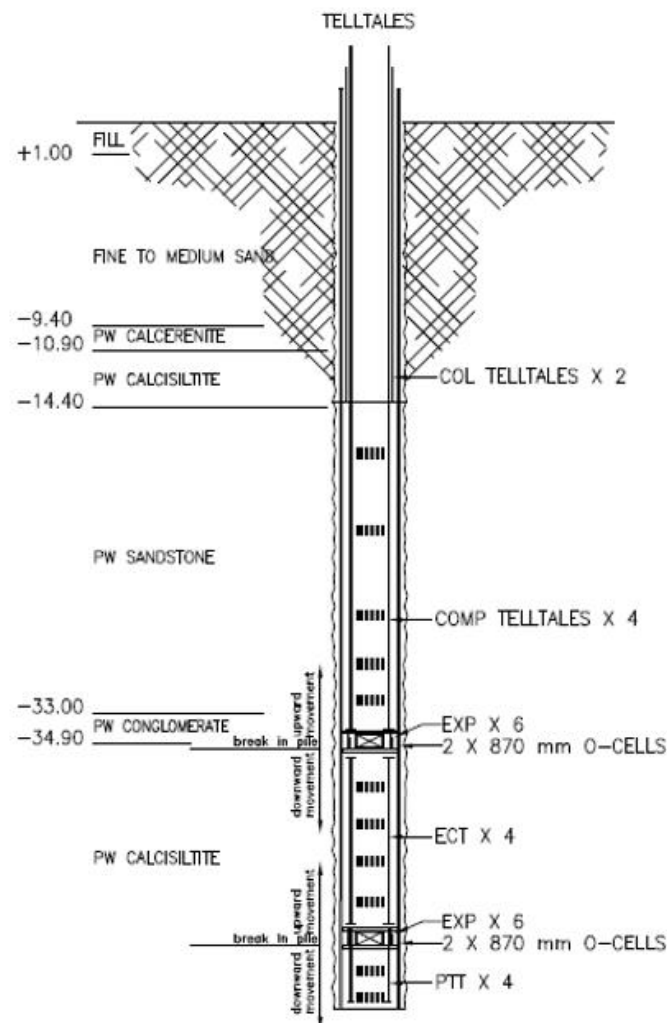


Figure 1: Pile-load test configuration: Stratigraphy and cell locations

The pile configuration under investigation is shown in Figure 1. Concrete was placed only between elevations -14.0 m and -51.3 m. A temporary steel casing was installed above -14.0 m to prevent shaft friction between the pile and the surrounding soil. The load testing was performed on a 1000 mm × 2800 mm barrette using two distinct O-cell assemblies positioned at different depths:

- The upper O-cell was installed between elevations -34.70 m and -35.20 m,
- The lower O-cell was positioned between -46.70 m and -47.20 m.

Displacements were recorded at both the upper and lower bearing plates for each O-cell assembly.

The load test was conducted in two stages:

- First, the lower O-cell assembly was pressurized to 53.20 MN and then unloaded.
- Next, the upper assembly was pressurized to 58.63 MN and subsequently relaxed. During this phase, the lower assembly was initially drained and then locked.

The experimental results from these tests are presented in Figure 2.

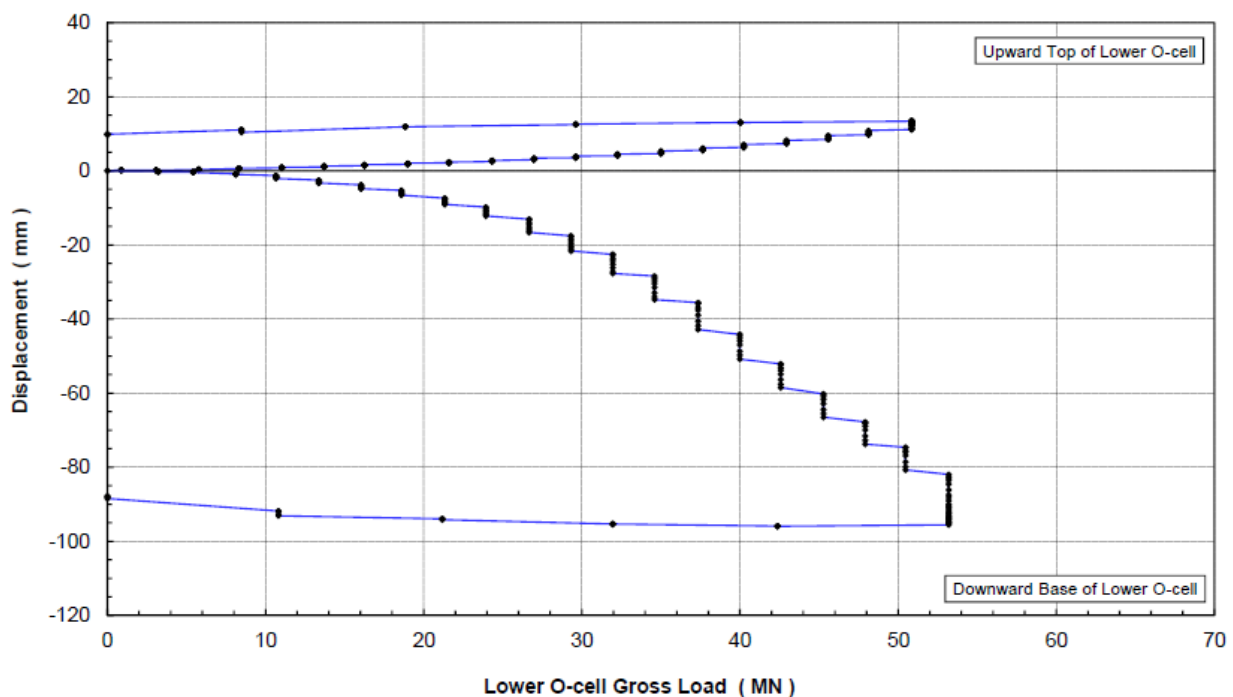


Figure 2: Pile-load test experimental results

This example focuses on modelling an O-cell pile load test in PLAXIS 3D using volume (continuum) elements. It highlights several modelling techniques aimed at:

- enhancing the computational efficiency of the numerical model, and
- achieving accurate calibration against the target load–displacement curves.

Taking advantage of the project's geometric symmetry, only one quarter of the barrette is modelled. Furthermore, the current analysis is limited to the first loading and unloading stage.

Keywords: PLAXIS 3D, Pile barrette, Volume pile, Osterberg cell test, Pile load test, Calibration, Experimental results.

2.

Model Construction

Start a new project and define the appropriate Contour dimensions according to the size of the geometry (see Figure 3).

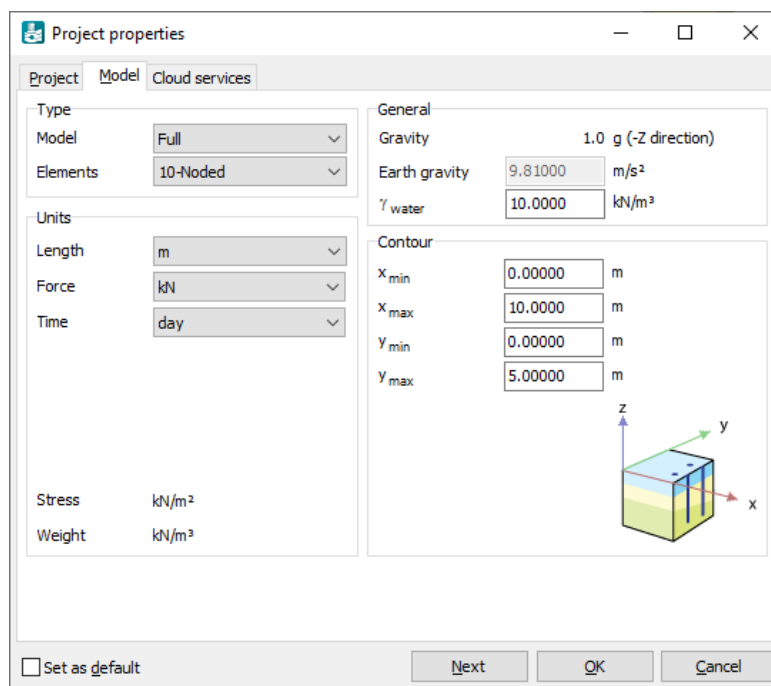


Figure 3: Model dimension definition in Project properties

The dimensions $x_{\max} = 10$ m and $y_{\max} = 5$ m have been chosen considerably larger than the barrette horizontal dimensions (of respectively 1.4 m and 0.5 m of the quarter being effectively modelled) in order to avoid boundary effects.

Soil mode

First create the soil material properties that will be used in this project as summarized in Table 1.

Click the create borehole button in the side (vertical) toolbar to start defining the soil stratigraphy and click at $x = 0$ in the drawing area to locate the borehole. The *Modify soil layers* window will appear as shown in Figure 4:

- Add 7 soil layers from 2.5 m to -9.4 m, -9.4 m to -10.9 m, -10.9 m to -14.4 m, -14.4 m to -34.9 m, -34.9 m to -75 m, -75 m to -117 m and -117 m to -130 m,
- Assign to each respective layer the soil material set parameters as defined in Table 1,
- Set water level head to -8 m.

Table 1: Soil material set parameters

Parameter	Silty Sand	Calcarenite	Calcisiltite/ Siltstone	Mudstone
Soil model	Mohr-Coulomb	Mohr-Coulomb	Mohr-Coulomb	Mohr-Coulomb
Drainage type	Drained	Drained	Drained	Drained
Unsaturated weight γ_{unsat}	18 kN/m ³	19 kN/m ³	19 kN/m ³	19 kN/m ³
Saturated weight γ_{sat}	18 kN/m ³	19 kN/m ³	19 kN/m ³	19 kN/m ³
<u>Mechanical properties</u>				
Young's modulus E_{ref}	30E3 kPa	300E3 kPa	1.7E6	1.7E6
Poisson's ratio ν	0.3	0.25	0.25	0.25
Cohesion c'_{ref}	1 kPa	35 kPa	38 kPa	38 kPa
Friction angle ϕ'	30°	32°	27°	27°
Dilatancy angle ψ	0°	0°	0°	0°
Tensile strength f_t	0 kPa	0 kPa	0 kPa	0 kPa
<u>Initial stress properties</u>				
Initial stress ratio K_0	Auto	Auto	Auto	Auto

Parameter	Sandstone	Calcisiltite
Soil model	HSSmall	HSSmall
Drainage type	Drained	Drained
Unsaturated weight γ_{unsat}	19 kN/m ³	20 kN/m ³
Saturated weight γ_{sat}	19 kN/m ³	20 kN/m ³
<u>Mechanical properties</u>		
Ref. secant modulus E_{50}^{ref}	200E3 kPa	500E3 kPa
Ref. oedometer modulus $E_{\text{oed}}^{\text{ref}}$	200E3 kPa	500E3 kPa
Ref. Young's modulus $E_{\text{ur}}^{\text{ref}}$	600E3 kPa	1.5E6 kPa
Poisson's ratio ν_{ur}	0.1	0.1
Power coefficient m	0	0
Reference pressure p_{ref}	100 kPa	100 kPa
Ref. dynamic shear modulus G_0^{ref}	750E3 kPa	1.5E6 kPa
Shear Strain 70%	0.4E-3	0.4E-3
Cohesion c'_{ref}	25 kPa	40 kPa
Friction angle ϕ'	35°	38°
Dilatancy angle ψ	0°	0°
Tensile strength f_t	0 kPa	0 kPa
<u>Initial stress properties</u>		
Initial stress ratio K_0	Auto	Auto

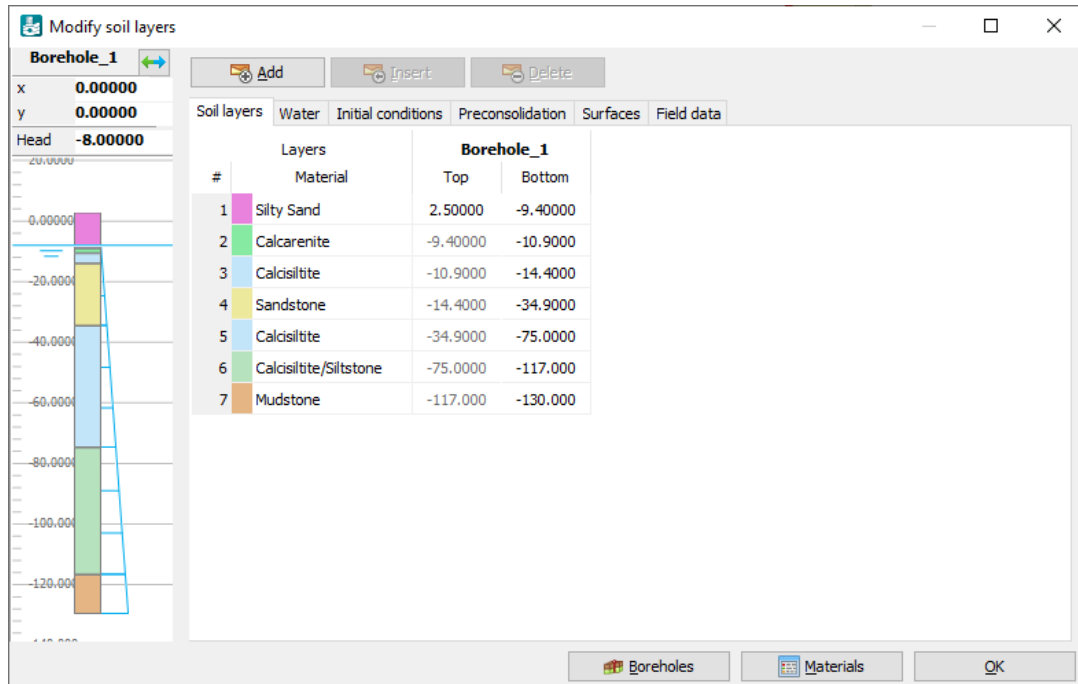


Figure 4: Borehole definition

Structures mode

Start defining a surface at top model level by creating a polygon using the following surface points:

- (0, 0, 2.5),
- (1.4, 0, 2.5),
- (1.4, 0.5, 2.5),
- (0, 0.5, 2.5).

Select the previously created polygon *Polygon_1* and extrude it downwards by an extrusion vector of (0, 0, -54.3 m).

Consider the previously created volume *Volume_1* and from the right-mouse click context menu select *Decompose into surfaces*. Select the outmost surfaces in X and Y direction and select *Create positive interface*.

We will also at this stage *Create prescribed displacement* along the same surfaces, the purpose of which will be later to maintain the soil in place once the soil will be excavated above the instrumented barrette. Displacement should be prescribed as such:

- Along surface at $y = 0.5$ m:
 - Displacement_x: Free,
 - Displacement_y: Fixed,
 - Displacement_z: Free.
- Along surface at $x = 1.4$ m:
 - Displacement_x: Fixed,
 - Displacement_y: Free,
 - Displacement_z: Free.

The initial polygon *Polygon_1* will now be copied multiple times to represent the top and bottom position of the barrette as well as the o-cell locations. Select *Polygon_1* and create arrays of "1D, in z

direction" shape with number of columns set to 2 each time with the following distance between columns:

- -16.5 m (barrette top position),
- -49.2 m (o-cell upper level),
- -49.7 m (o-cell lower level),
- -53.8 m (barrette bottom position).

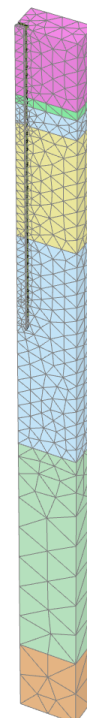
This will result in the creation of 4 additional surfaces respectively named *Polygon_2*, *Polygon_3*, *Polygon_4* and *Polygon_5* at the following respective elevations: -14 m, -46.7 m, -47.2 m and -51.3 m.

Create negative interface on *Polygon_5* corresponding to the barrette bottom surface.

The pressure load generated by the o-cell will be generated by creating surface loads on *Polygon_3* and *Polygon_4* respectively. Finally set the pressure to $\sigma_z = 19000$ kPa for the o-cell upper level (*Polygon_3*) and $\sigma_z = -19000$ kPa for the o-cell lower level (*Polygon_4*).



(a) Mesh refinement



(b) Generated mesh

Figure 5: Mesh definition

Mesh mode

Select the five top clusters surrounding the barrette lateral outer contour and set their respective coarseness factor to 0.4 (see Figure 5(a)).

Click on *Generate mesh* and set the *Element distribution* to *Medium*. The generated mesh is presented in Figure 5(b).

Staged construction mode

Go to the *Staged construction* mode. Start creating the material property sets for the concrete materials as well as the barrette/soil interface properties about to be activated as given in Table 2 and Table 3 respectively.

Table 2: Concrete material properties

Material Name	Concrete	Concrete Low Stiffness
Soil model	Linear Elastic	Linear Elastic
Unit weight unsaturated γ_{unsat}	24 kN.m ⁻³	24 kN.m ⁻³
Drainage type	Non-porous	Non-porous
<u>Mechanical properties</u>		
Young modulus E_{ref}	20E6 kPa	20 kPa
Poisson ratio ν	0.2	0.2

Table 3: Interface material set parameters

Parameter	Interface Sandstone	Interface Calcisiltite
Soil model	Mohr-Coulomb	Mohr-Coulomb
Drainage type	Drained	Drained
Unsaturated weight γ_{unsat}	19 kN/m ³	20 kN/m ³
Saturated weight γ_{sat}	19 kN/m ³	20 kN/m ³
<u>Mechanical properties</u>		
Young's modulus E_{ref}	600E3	1.5E6
Poisson's ratio ν	0.25	0.25
Cohesion c'_{ref}	225 kPa	400 kPa
Friction angle ϕ'	0°	0°
Dilatancy angle ψ	0°	0°
Tensile strength f_t	0 kPa	0 kPa

Initial conditions

For the initial phase, K0-procedure will be used which is the default phase type in PLAXIS so no specific action is required at that stage.

Phase 1: Excavation

Create a first phase and name it *Pile Installation*:

- Deactivate the top 3 soil clusters above the barrette (from $z = -14$ m upwards) and set their *WaterConditions* to Dry.
- Activate prescribed displacement along the 6 cut faces to maintain remaining soil in place (see Figure 6).
- Select the 5 soil clusters corresponding to the barrette (from $z = -14$ m down to $z = -51.3$ m) and set their material to *Concrete* as defined in **Error! Reference source not found..** Set their *WaterConditions* to *Dry* as well.
- Activate all interface elements around and underneath the barrette volume with the addition of the interface extensions from $z = -51.3$ m to $z = -51.8$ m.
- Assign proper interface properties depending on whether the barrette is contacting the calcisiltite or the sandstone as shown in Figure 7.

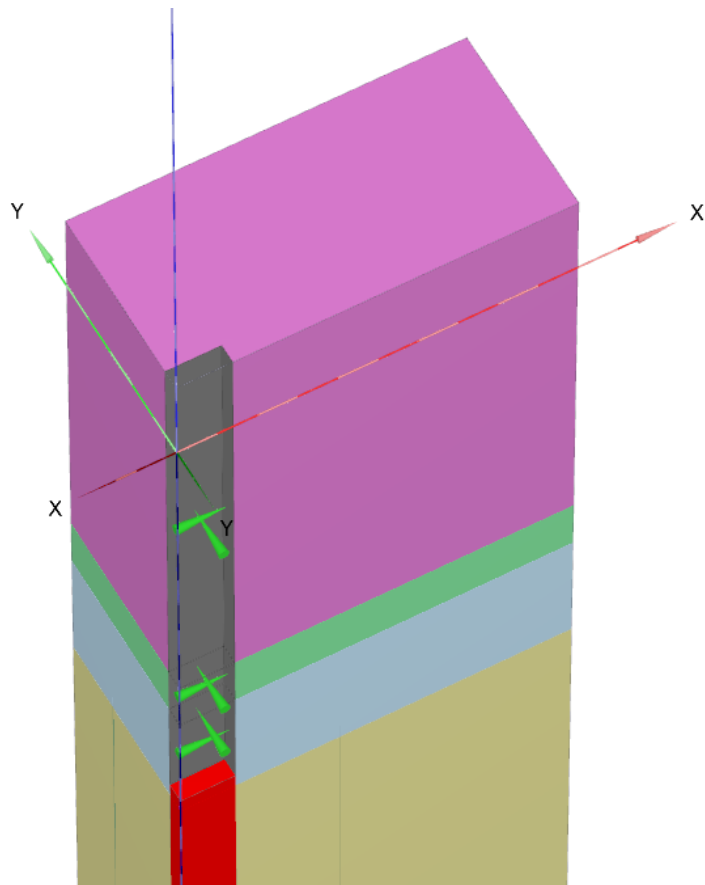


Figure 6: Activation of prescribed displacement



Figure 7: Barrette/surrounding soil interface definition

Phase 2: Loading

Create a second phase entitled *Loading*:

- Start with activating *Reset displacements to zero* from the Phase settings dialog box.
- Assign o-cell cluster (from $z = -46.7$ m to $z = -47.2$ m) with *Concrete low stiffness* material set.
- Activate surface loads

Phase 3: Unloading

Create a third phase entitled *Unloading*:

- Deactivate surface loads

Before starting the calculation process, two nodes belonging to respectively the upper and lower o-cell parts must be selected for being able to display the characteristic load-displacement curves

- Click the Select points for curves button. The model and Select points window will be displayed in the Output program.
- Define (0, 0, -47.2) and (0, 0, -46.7) as Point-of-interest coordinates

The calculation can be launched by clicking on *Calculate*.

3.

Main Calculation Results

Once the calculation has completed, one can open the *Output* program.

Load-displacement curves

Select the *Curves manager* tool and click on *New* in the *Charts* tabsheet to create a new curve. In the *Curve generation* dialog box, select the ΣM_{stage} Project result as a *X-Axis* result and the total displacement u_z for the lower o-cell part node at (0, 0, -47.2) as the *Y-Axis* result (see Figure 8).

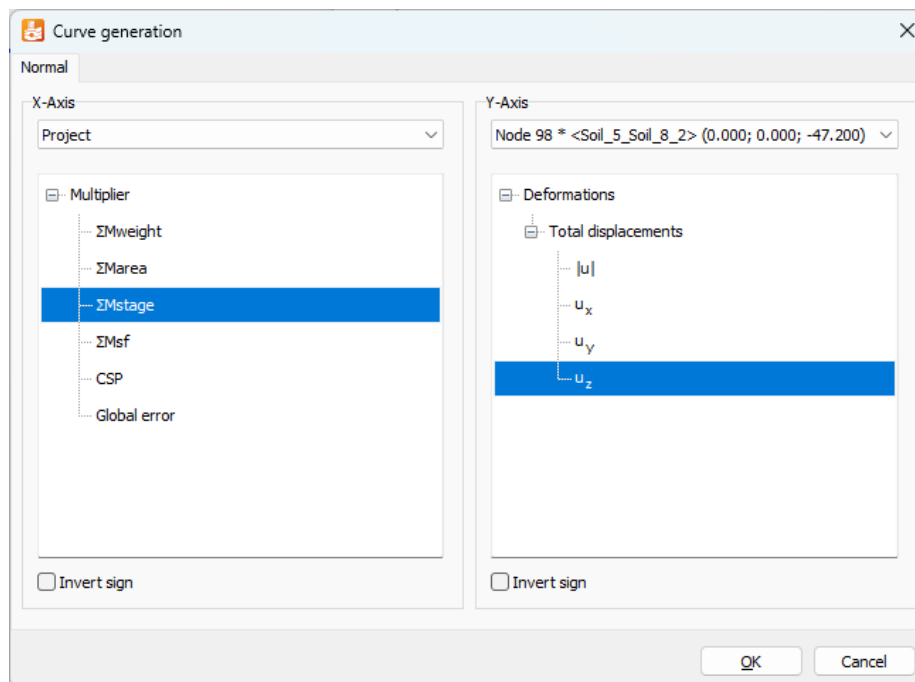


Figure 8: Selection of axis attributes for curve generation

From the *Curves Settings* only select last phase (*Vertical Loading [Phase 2]*) calculation results to be displayed as shown in Figure 9.

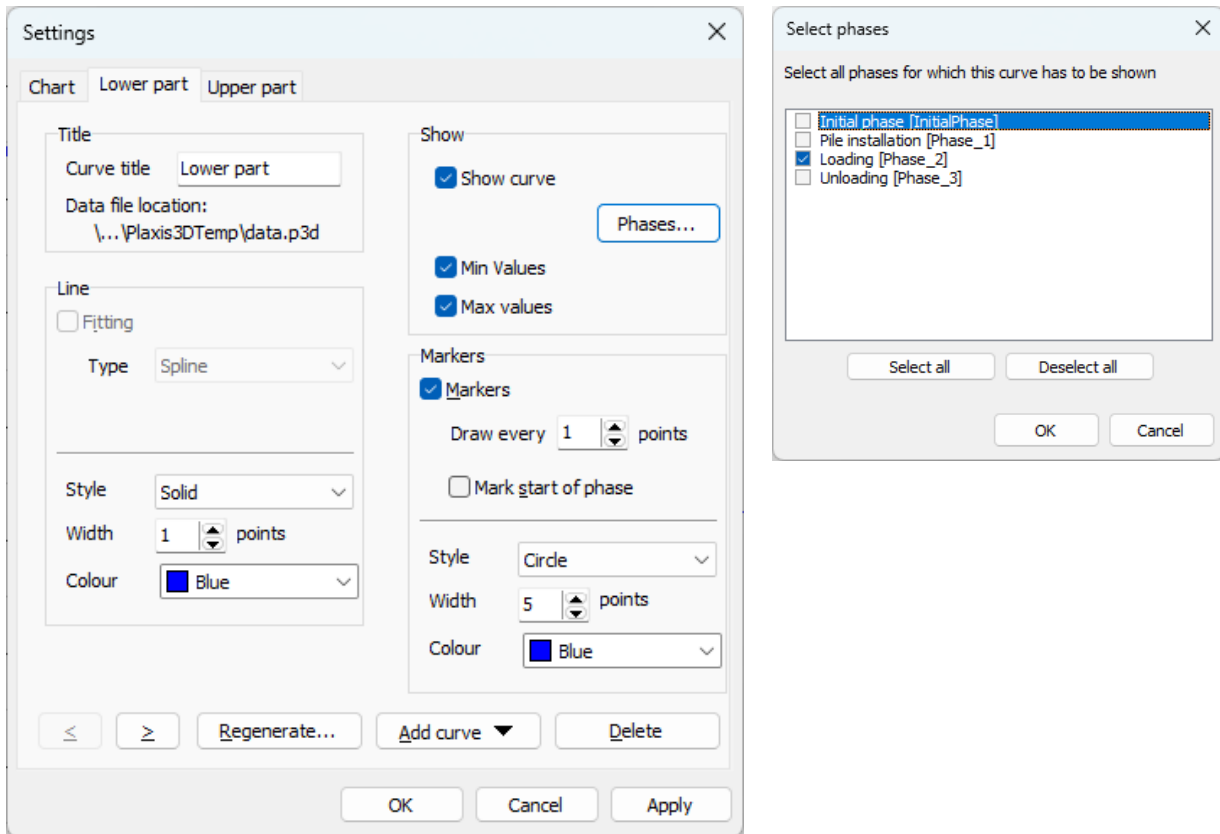


Figure 9: Curve settings for phase selection

Repeat these operations for upper o-cell part node at (0, 0, -46.7). Generated load-displacement curves at both pile top and bottom locations are finally shown in Figure 10.

The characteristic load-displacement curves obtained numerically and how they would compare against the experimental values can be plotted using the **postProcessOsterberg.py** script given in Appendix.

Running the **postProcessOsterberg.py** script will display both the numerical and experimental load-displacement curves as shown in Figure 11. One can easily check optimal match of numerical results against experimental results for both upper and lower cell parts.

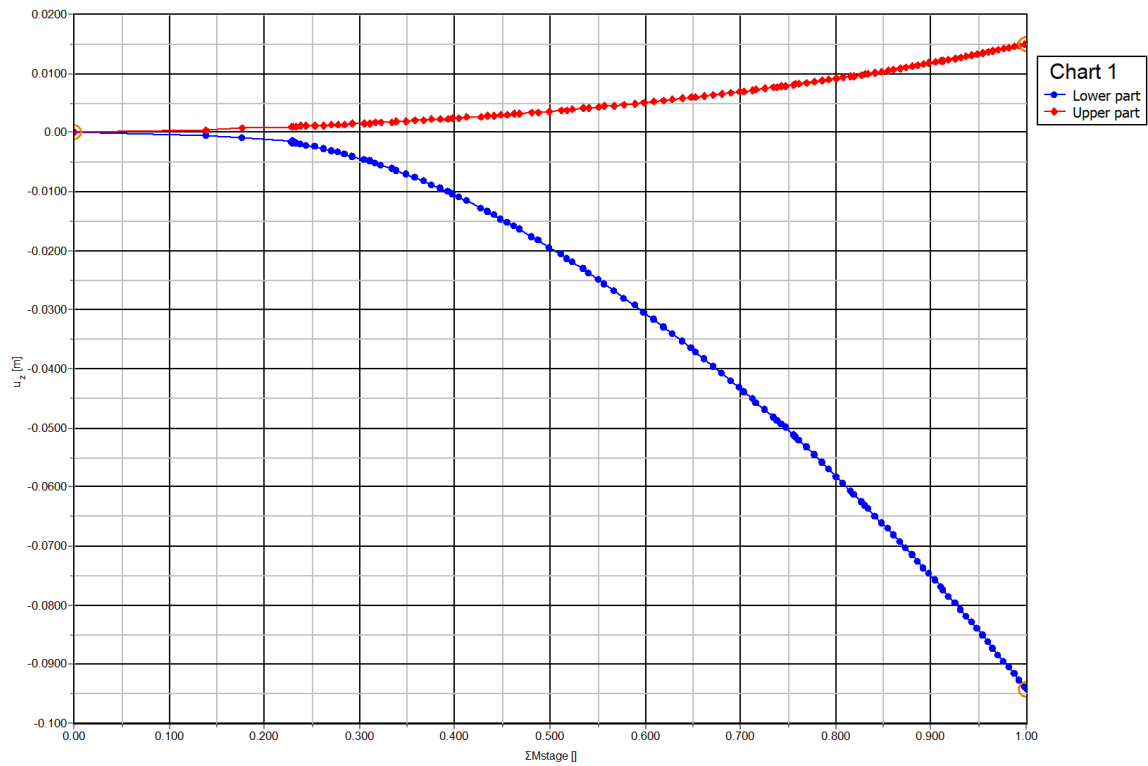


Figure 10: Calculated load-displacement curves at both pile top and bottom locations

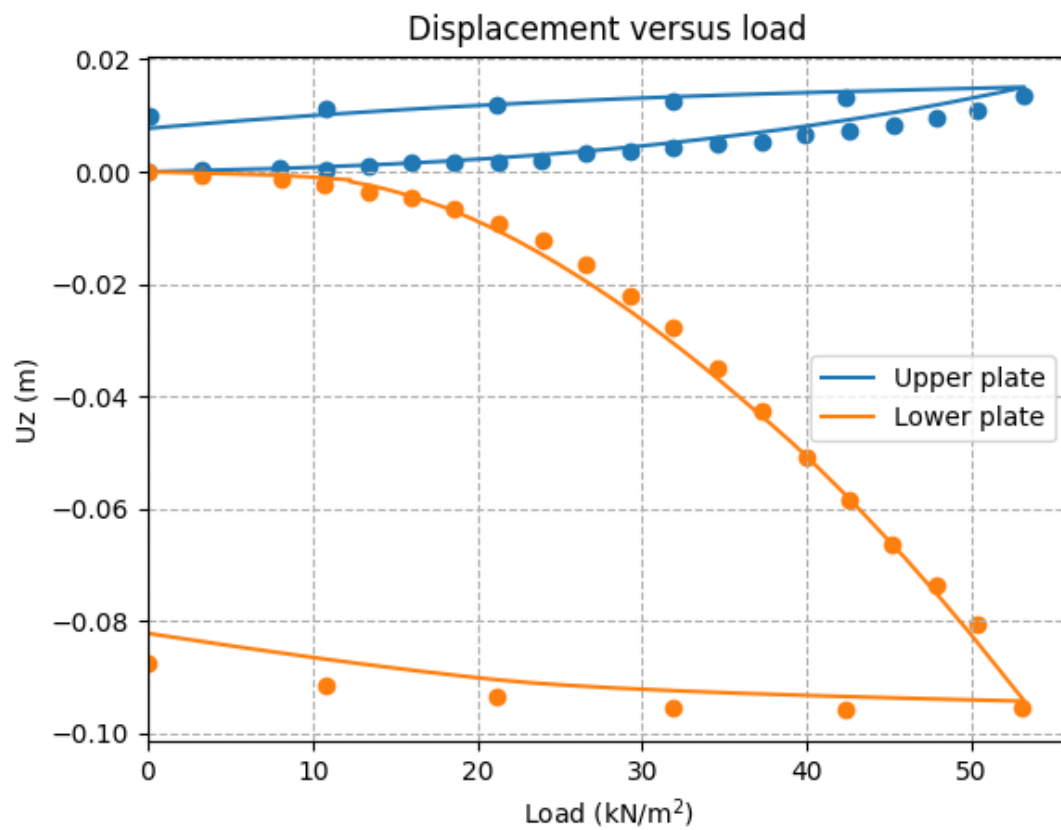


Figure 11: Load-displacement curves

4.

Sensitivity Analysis

The material parameters provided in Table 1 have been initially calibrated such that optimal match against experimental results is obtained and that user running this example can immediately experience an optimized set of material properties. It is however crucial to realize how important the effect of properly considering the appropriate soil stiffness to obtain proper match.

In this context, we are proposing to reconsider the numerical analysis of the Osterberg test using Mohr-Coulomb constitutive model for the sandstone and the calcisiltite layers as shown in Table 4. When using MC model, the effect of strain level on the soil shear stiffness is not accounted for and therefore no stiffness degradation mechanism is taking place. The soil stiffness remains at its initial user input value $G_{0,ref}$ and v_{ur} .

Table 4: Alternative soil material set parameters for the sandstone and calcisiltite layers

Parameter	Sandstone MC	Calcisiltite MC
Soil model	Mohr-Coulomb	Mohr-Coulomb
Drainage type	Drained	Drained
Unsaturated weight γ_{unsat}	19 kN/m ³	20 kN/m ³
Saturated weight γ_{sat}	19 kN/m ³	20 kN/m ³
<u>Mechanical properties</u>		
Young's modulus E_{ref}	1.65E6 kPa	3.9E6 kPa
Poisson's ratio ν	0.1	0.1
Shear Modulus G_{ref}	750E3 kPa	1.5E6 kPa
Cohesion c'_{ref}	25 kPa	40 kPa
Friction angle ϕ'	35°	38°
Dilatancy angle ψ	0°	0°
Tensile strength f_t	0 kPa	0 kPa
<u>Initial stress properties</u>		
Initial stress ratio K_0	Auto	Auto

In the *Soil* mode, create the two additional sets of materials parameters namely and replace the corresponding material in *Borehole_1* (as shown in Figure 12). Then rerun the analysis and the script **postProcessOsterberg.py** in *Output* once the new analysis has been completed. It will display both the experimental and updated numerical load-displacement curves as shown in Figure 13 where a clearly over-stiff numerical answer for the lower o-cell part can now be observed.

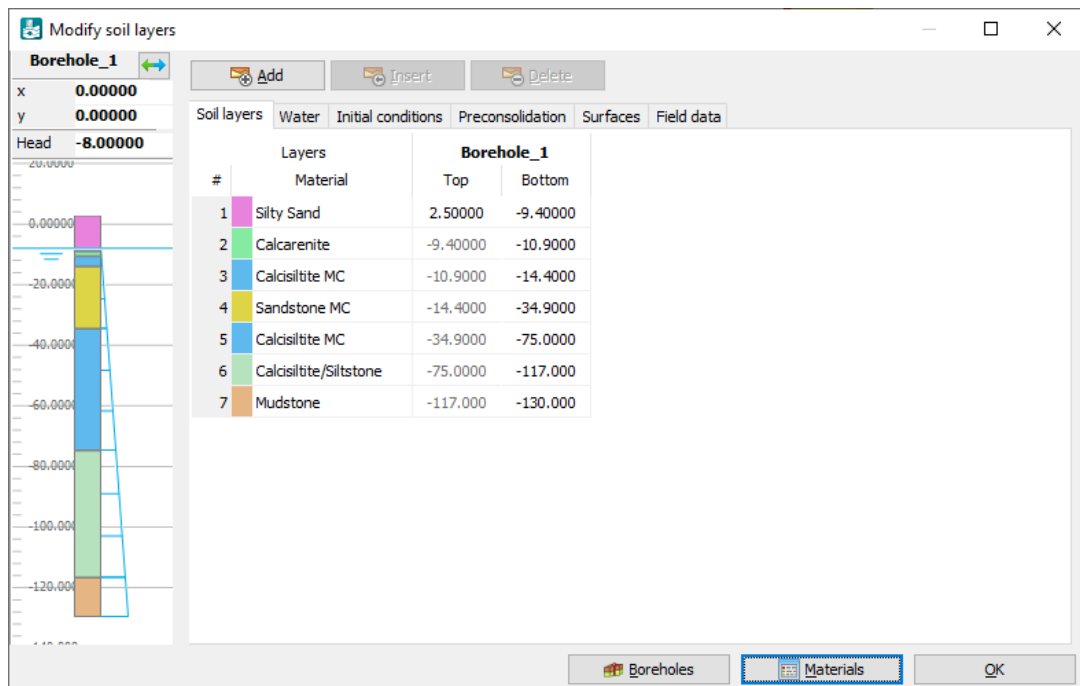


Figure 12: Updated borehole definition for sensitivity analysis

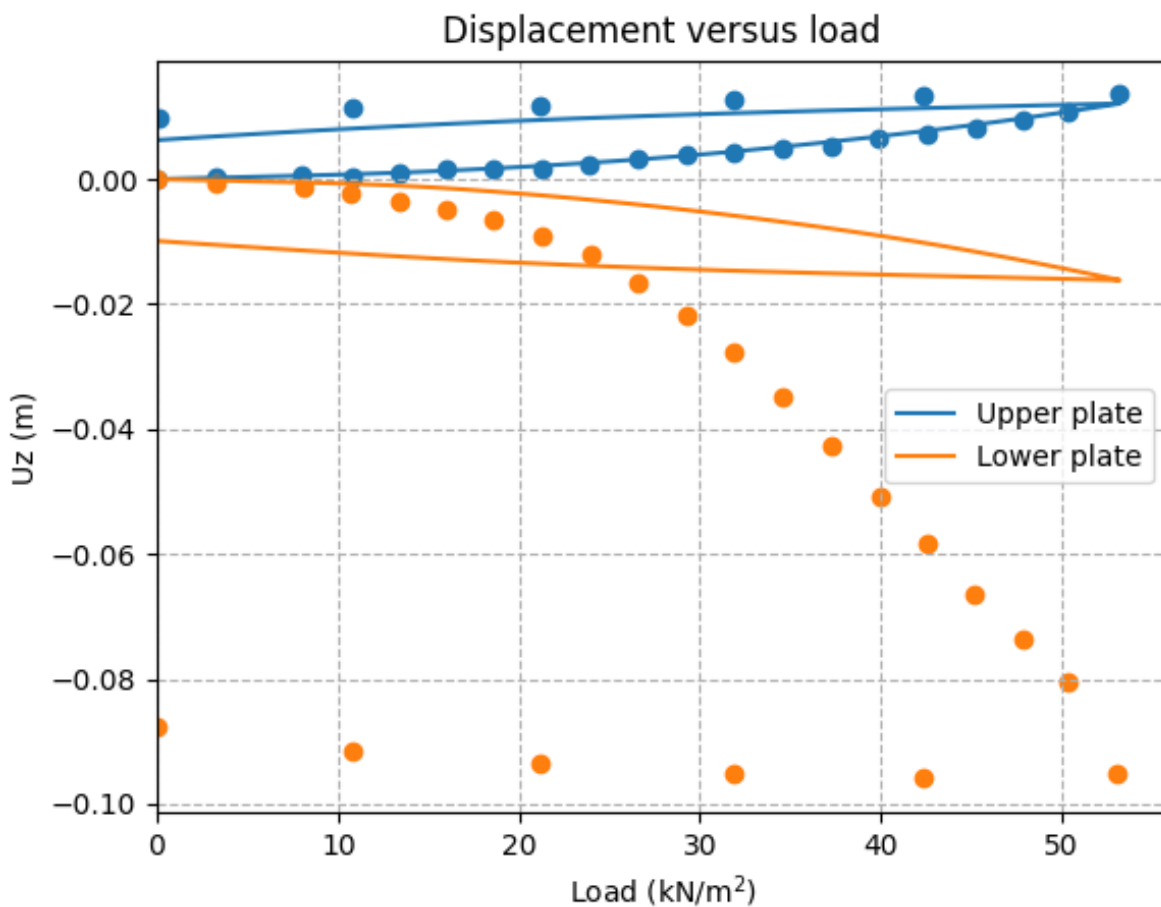


Figure 13: Load-displacement curves obtained using MC constitutive model

5.

Appendix

The python code provided in **postProcessOsterberg.py** can be used to generate the characteristic load-displacement curves obtained numerically and how they would compare against the experimental values:

postProcessOsterberg.py

```
from plxscripting.easy import *
import matplotlib.pyplot as plt

# Initialize input scripting server
s_o, g_o = new_server()
#
maxLoad = 53.2
phaseLoading = g_o.Phases[-2]
phaseUnloading = g_o.Phases[-1]
nodeTop = g_o.Nodes[1]
nodeBottom = g_o.Nodes[0]
#
loadNum = [0]
upperPlateNumDisp = [0]
lowerPlateNumDisp = [0]
upperPlateExpLoad = [0, 3.27291769148986,
8.02177121185674, 10.7810342912454,
13.3481648391754, 15.9793713007188,
18.5463114298862, 21.3056697186562,
23.8727050572049, 26.5683682698868,
29.2638410638061, 31.8308764023548,
34.5905203192687, 37.2858979038067,
39.9174852028753, 42.5488820831812,
45.3086212094765, 47.811961471937,
50.3793776480109, 53.1399736587378,
42.2949586632602, 31.8989511099827,
21.1819927323505, 10.7859851790729,
0.0687411732969418]
upperPlateExpDisp = [5.93471810088956E-05, 0.000267062314540055,
0.000682492581602371, 0.000474777448071211,
0.00109792284866468, 0.001513353115727,
0.00172106824925815, 0.00172106824925815,
0.00213649851632047, 0.00317507418397625,
0.00379821958456972, 0.00421364985163204,
0.00483679525222551, 0.00525222551928783,
0.00649851632047477, 0.00732937685459939,
0.00816023738872403, 0.00961424332344213,
0.010860534124629, 0.0135608308605341,
0.0133531157270029, 0.0127299703264094,
0.0118991097922848, 0.0112759643916913,
0.00982195845697328]
lowerPlateExpLoad = [0, 3.27253685396467,
8.08508545041971, 10.7156254462939,
13.3460702327869, 15.9765150192799,
18.5425030546343, 21.3007188308288,
23.9944778558847, 26.6234945016582,
29.2520351005252, 31.8804804900109,
34.5723353274409, 37.3280756597216,
40.0195496596264, 42.5187959186911,
45.2102699185959, 47.8379536330313,
50.3374855202399, 53.0900839429378,
42.3090496516923, 31.8494422317079,
21.1977974896459, 10.8030276583252,
0]
lowerPlateExpDisp = [0, -0.000563798219584573,
-0.00118694362017804, -0.00222551928783383,
-0.00347181008902077, -0.00471810089020771,
-0.00658753709198813, -0.00908011869436202,
-0.0121958456973293, -0.0165578635014836,
-0.0219584569732937, -0.027566765578635,
-0.0348367952522255, -0.0427299703264095,
-0.0508308605341246, -0.0583086053412462,
-0.0664094955489614, -0.0736795252225519,
-0.0805341246290801, -0.0952818991097922,
-0.0959050445103857, -0.0952818991097922,
-0.093620178041543, -0.0915430267062314,
-0.0875964391691394]
```

```

# Fetch results
unitLength = g_o.GeneralInfo.UnitLength.value
unitForce = g_o.GeneralInfo.UnitForce.value
loadNum = (loadNum +
            [step.Reached.SumMstage.value * maxLoad for step in phaseLoading.Steps.value] +
            [(1 - step.Reached.SumMstage.value) * maxLoad for step in phaseUnloading.Steps.value])
result = g_o.ResultTypes.Soil.Uz
upperPlateNumDisp = (upperPlateNumDisp +
                     [g_o.getcurveresults(nodeTop, step, result) for step in phaseLoading.Steps.value] +
                     [g_o.getcurveresults(nodeTop, step, result) for step in phaseUnloading.Steps.value])
lowerPlateNumDisp = (lowerPlateNumDisp +
                     [g_o.getcurveresults(nodeBottom, step, result) for step in phaseLoading.Steps.value] +
                     [g_o.getcurveresults(nodeBottom, step, result) for step in phaseUnloading.Steps.value])

# Plot results
fig, ax = plt.subplots()
ax.set_title('Displacement versus load')
ax.set_xlabel("Load (" + unitForce + "/" + unitLength + "$^2$)")
ax.set_ylabel("Uz (" + unitLength + ")")
ax.scatter(upperPlateExpLoad, upperPlateExpDisp)
ax.scatter(lowerPlateExpLoad, lowerPlateExpDisp)
ax.plot(loadNum, upperPlateNumDisp, label="Upper plate")
ax.plot(loadNum, lowerPlateNumDisp, label="Lower plate")
ax.legend()
plt.xlim(left=0)
plt.grid(True, ls='--')
plt.show()

```